Chronic Kidney Disease in Zimbabwe

How common it is, and practical ways to improve life for people living with it



What CKD actually is

DEFINITION

Abnormal kidney structure or function lasting more than 3 months

GFR below 60 mL/min/1.73 m² and/or markers of damage such as albuminuria

Staged G1 to G5. G5 (GFR below 15) is kidney failure.

The kidney has seven jobs. Lose the job, get the problem.

Waste removal	→	Uraemia
Salt and water balance	→	Oedema, hypertension
Acid removal	→	Metabolic acidosis
Potassium excretion	→	Hyperkalaemia
Phosphate excretion	→	Hyperphosphataemia
Vitamin D activation	→	Low calcium, bone disease
Erythropoietin (EPO)	→	Anaemia

How common is CKD?

13.7%

Stages 1 to 5, pooled across community studies in Africa (2026 meta-analysis)

5.1%

Stages 3 to 5, the advanced range with real clinical consequences (2026)

15.8%

2018 meta-analysis, stages 1 to 5, 22 general-population studies. A 2014 review found 13.9%.

10.7%

Adults aged 40 to 60 in a study of over 8,000 people across four African countries (2019)

Zimbabwe-specific population data are limited, so regional figures are the best available proxy. They fit what we see on the wards: CKD alongside diabetes, hypertension, HIV and amyloidosis.

Sources: George et al., Am J Kidney Dis 2026; Kaze et al., BMC Nephrol 2018; George et al. (AWI-Gen), Lancet Glob Health 2019; Stanifer et al., Lancet Glob Health 2014. Full references at the end.

What drives CKD here

Hypertension

Hardens the small afferent arterioles, so glomeruli become ischaemic and scar. Kidney damage then raises BP further.

Diabetes

High glucose raises pressure inside the glomerulus. The filter leaks albumin and scars.

HIV

Infects podocytes and causes collapsing FSGS (HIVAN). APOL1 variants raise risk. Tenofovir can injure tubules.

Infection and obstruction

TB, schistosomiasis, stones, BPH and cervical cancer scar or block the urinary tract.

Glomerular and systemic disease

Glomerulonephritis, lupus, myeloma and amyloidosis (AA type follows chronic infection or inflammation).

Nephrotoxic drugs

Long-term NSAIDs and other nephrotoxins reduce renal blood flow and injure the interstitium.

Why it keeps getting worse

Nephrons are lost

Any cause: diabetes, hypertension, HIV, infection, drugs

Survivors overwork

Afferent arteriole dilates: more flow and pressure per glomerulus (hyperfiltration)

The filter is damaged

Podocytes stretch and detach, protein leaks, glomeruli scar

Even fewer nephrons

GFR falls and the remaining nephrons work harder still

The cycle repeats, even after the original cause has been removed

- Leaked protein is itself toxic to the tubules, so proteinuria is both a marker and a driver of progression.
- That is why blood pressure and proteinuria are treated early, not just the original disease.

What CKD does to a life

Physical	Fatigue and breathlessness (anaemia, fluid overload), itching, nausea, poor appetite, swelling.
Treatment burden	Long trips to dialysis centres, hours on a machine several times a week, strict diet and fluid limits.
Financial	Dialysis and specialist care are costly and not always accessible outside major cities. Families face catastrophic costs and lost income.
Psychosocial	Anxiety, depression, loss of independence and work, and strain on relationships and family roles.

Solutions: five levers

1

Find it early

Screen high-risk groups with urine protein and creatinine

2

Slow the decline

Control BP, sugar and infection; protect the kidney

3

Treat complications

Anaemia, fluid, acid, potassium, bone disease

4

Access to care

Dialysis closer to home, transplant, conservative care

5

Support the person

Education, mental health, peer groups, financial help

Health-system change underpins all five: medicines, training, financing and a CKD registry

1. Find it early

Who: people with hypertension, diabetes or HIV, a family history of kidney disease, chronic infection such as TB, or long-term NSAID use.

How: urine dipstick or albumin-creatinine ratio, plus serum creatinine to estimate GFR. Both are cheap and already available.

Where: add it to HIV, hypertension and diabetes clinics patients already attend. One extra test, not a new programme.

By whom: nurses and primary care staff trained to screen, interpret and refer.



Why it matters

Early CKD is silent. By the time a patient feels unwell, much of the kidney function may already be gone.

2. Slow the decline

Control the drivers

- Blood pressure: aim for below about 130/80
- ACE inhibitor or ARB when there is proteinuria: lowers pressure inside the glomerulus
- Glycaemic control in diabetes
- Antiretroviral therapy for HIV; treat TB and other infections
- Relieve obstruction: stones, BPH, strictures

Protect the kidney

- Avoid NSAIDs and other nephrotoxins; adjust drug doses to eGFR
- Less salt, regular activity, no smoking
- Check potassium and creatinine after starting an ACE inhibitor or ARB
- SGLT2 inhibitors slow progression in many patients, where affordable
- Monitor GFR and urine protein regularly

3. Treat the complications

Anaemia	Check and replace iron; erythropoiesis-stimulating agents (EPO) where available; look for other causes such as bleeding or B12 or folate deficiency.
Fluid and blood pressure	Salt restriction and diuretics; dialysis if overload does not respond.
Acidosis	Oral sodium bicarbonate to correct low bicarbonate.
High potassium	Limit dietary potassium, review drugs that raise it, treat severe rises as an emergency.
Bone and mineral disease	Control phosphate (diet and binders); manage calcium and vitamin D.

Each maps to a lost kidney function: EPO, acid, potassium, phosphate, vitamin D, salt and water.

4. Access to kidney care

Dialysis closer to home

- Peritoneal dialysis needs less infrastructure and can be done at home after training
- Satellite haemodialysis units beyond the main cities
- Transport and cost support for patients on treatment

Transplantation

- Best long-term survival and quality of life for suitable patients
- Needs donor pathways and a steady supply of immunosuppressants
- Early referral and work-up

Conservative kidney care

- For patients who cannot access or do not want dialysis
- Symptom control for itch, nausea, breathlessness and pain
- Palliative care and advance care planning
- A legitimate choice, not "no treatment"

5. Support the whole person

Education and self-management

Teach diet, fluid limits, medicines and warning signs. Nurse and dietitian-led, in English, Shona and Ndebele, with simple picture-based materials.

Mental health

Screen routinely for depression and anxiety. Offer counselling and link to community mental-health services.

Peer and community support

Patient support groups, community health workers, and church and community networks that help with transport, meals and encouragement.

Financial and social protection

Subsidised medicines and dialysis, insurance coverage, transport support, and flexible work or school arrangements.

Health-system changes needed

START WITH THE DATA GAP

Zimbabwe has little population-level CKD data. A national CKD registry and screening data from NCD and HIV clinics would show the true burden and guide funding.

1. **National CKD strategy:** built into existing NCD and HIV programmes
2. **Reliable essential medicines:** ACE inhibitors, iron, bicarbonate and other core CKD drugs stocked consistently
3. **Decentralised expertise:** train district nurses and doctors; link them to nephrologists by telemedicine
4. **Financing:** insurance or subsidy schemes for dialysis and transplant, so cost does not decide survival
5. **Prevention upstream:** better hypertension, diabetes and HIV control means less CKD later

Take-home messages

1. About 1 in 7 adults in pooled African studies has CKD (13.7% in the latest 2026 analysis). Zimbabwe-specific data are limited and urgently needed.

2. Hypertension, diabetes, HIV, infections, obstruction and nephrotoxic drugs drive most disease.

3. Once nephrons are lost, hyperfiltration keeps the damage going, so early action matters most.

4. Quality of life improves when we find it early, slow the decline, treat complications, widen access to care and support the whole person.

Thank you. Questions?

References

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